

XIN BAI

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EDUCATION

University of Utah

Salt Lake City, UT

B.S. in Electrical Computer Engineering

12/2025

TECHNICAL SKILLS

SolidWorks	Python/C++	PLC	SLAM	OrCAD	Sensor
EDA:Pspice/LTspice	Digital System Design 3D	MATLAB	PCB	ArduPilot(APM)	PX4
Python	COMSOL	AutoCAD	Linux	Gazebo	ROS

PROFESSIONAL EXPERIENCE

Quicktron Automation

Hybrid

Project Engineer

07/2026

- **Supported end-to-end AGV/AMR project deployment**, including site infrastructure setup, system commissioning, UAT execution, production launch, and post-deployment maintenance to ensure stable warehouse automation operations.
- **Performed advanced AGV troubleshooting across software, network, electrical, and hardware layers**, using Final Shell to access Linux/ROS environments, retrieve and analyze system logs, verify network connectivity and device communication, and isolate root causes between PLC, safety LiDAR, controllers, communication modules, and AGV hardware.
- **Validated RCS and production-line integrations through API-based task dispatching and end-to-end workflow testing**, including transport-order creation, button-box functionality, station interaction, AGV task execution, and system status feedback during UAT and production support.
- **Configured and developed system nodes using LES Lite**, supporting workstation logic, equipment interfaces, task-flow configuration, and integration between RCS, AGVs, button boxes, and production-line equipment.
- **Developed strong hands-on knowledge of AGV architecture and field maintenance**, including safety LiDAR, PLC, CBox/DBox, motor controllers, drive motors, lifting mechanisms, batteries, sensors, communication networks, and related electrical components, enabling efficient fault diagnosis, component replacement, and recovery of production operations.

Hanseup Kim

University Of Utah

Hardware & Software Engineering (UAV Project)

01/2025

- Engineered a **robust UAV stabilization architecture** that improving controllability and safety by 35% in turbulent airflow conditions common to real-world urban and natural environments. Integrated GPS, IMU, and digital airspeed sensors (MATEKSYS ASPD-4525) with a Pixhawk flight controller.
- Implemented **end-to-end sensor integration** across GPS, IMU, and airspeed systems, improving state estimation fidelity and flight robustness.
- Designed and validated **real-time wind estimation and sensor-fusion algorithms** in MATLAB, directly informing adaptive control strategies.
- Delivered **low-level firmware and middleware integrations** within PX4 and ArduPilot, spanning build systems, cross-compilation, and custom sensor drivers for real-time flight control.

BYD

Guizhou,china

Facility Management Engineer

05/2023

- Improved **sensor positioning accuracy by 25%**, directly increasing marking-head precision and repeatability through calibration and hardware optimization.
- Performed **hardware-level inspections and diagnostics** on electrical and electromechanical systems supporting precision equipment, ensuring stable and compliant operating conditions for engineering hardware.
- Developed and executed **preventive maintenance plans** that improved 30% reliability of electrical and mechanical assets used in production and testing environments.
- Improved overall system throughput by **troubleshooting hardware issues, tuning system performance, and optimizing operational workflows**.